

# School of Computer Science Engineering and Technology

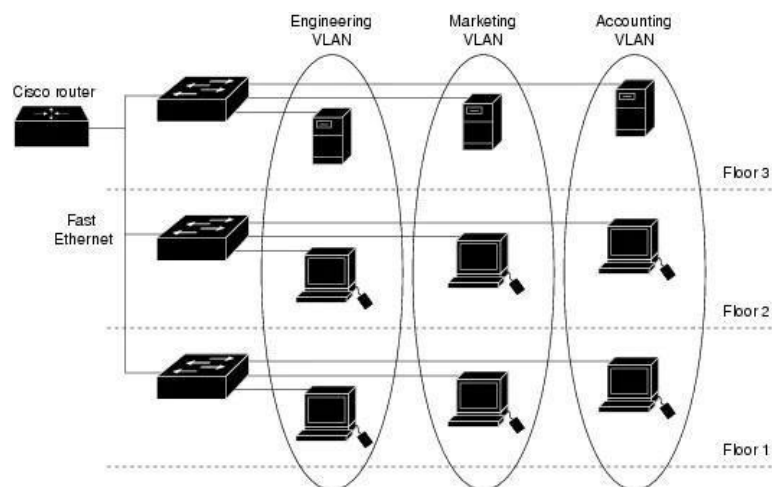
|                             |                                       |
|-----------------------------|---------------------------------------|
| <b>Course-</b> B. Tech      | <b>Type-</b> Core                     |
| <b>Course Code-</b> CSET207 | <b>Course Name-</b> COMPUTER NETWORKS |
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|           | <b>CO1</b> | <b>CO2</b> | <b>CO3</b> |
|-----------|------------|------------|------------|
| <b>Q1</b> | ✓          |            |            |
| <b>Q2</b> | ✓          |            |            |
| <b>Q3</b> | ✓          |            |            |
| <b>Q4</b> | ✓          | ✓          |            |

**Objective - Enterprise Switching: Understanding Virtual LAN (VLAN). VLAN Trunking Protocol (VTP), VLAN Tagging, Dynamic Trunking Protocol (DTP), Spanning Tree Protocol (STP), Router on the Stick (VLAN Routing).**

A **VLAN** is a group of end stations in a switched network that is logically segmented by function or application, without regard to the physical locations of the users. VLANs have the same attributes as physical LANs, but you can group end stations even if they are not physically located on the same LAN segment.

Any switch port can belong to a VLAN, and unicast, broadcast, and multicast packets are forwarded and flooded only to end stations in that VLAN. Each VLAN is considered as a logical network, and packets destined for stations that do not belong to the VLAN must be forwarded through a router. The following figure shows VLANs as logical networks. The stations in the engineering department are assigned to one VLAN, the stations in the marketing department are assigned to another VLAN, and the stations in the accounting department are assigned to another VLAN.

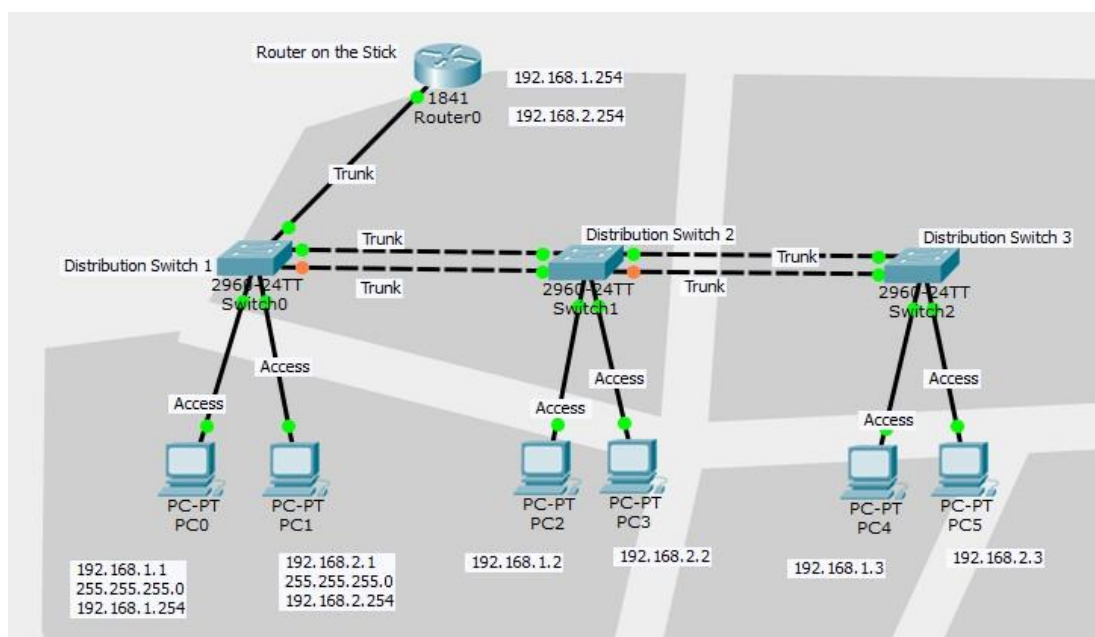


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| VLAN Number             | Range                | Usage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                       | Normal               | Cisco default. You can use this VLAN, but you cannot modify or delete it.                                                                                                                                                                                                                                                                                                                                                                                                             |
| 2—1005                  | Normal               | You can create, use, modify, and delete these VLANs.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1006—3967 and 4048—4093 | Extended             | You can create, name, and use these VLANs. You cannot change the following parameters: <ul style="list-style-type: none"> <li>The state is always active.</li> <li>The VLAN is always enabled. You cannot shut down these VLANs.</li> </ul>                                                                                                                                                                                                                                           |
| 3968-4047 and 4094      | Internally allocated | These 80 VLANs and VLAN 4094 are allocated for internal device use. You cannot create, delete, or modify any VLANs within the block reserved for internal use.                                                                                                                                                                                                                                                                                                                        |
| 3968-4094               | Internally allocated | Beginning with Cisco release 5.2(1) for Cisco Nexus 7000 Series devices, VLANs 3968 to 4094 are reserved for internal use in each VDC by default.<br><br>You can change the reserved VLANs to any other 128 contiguous VLAN range. When you reserve such a range, it frees up the range of VLANs that were allocated for internal use by default, and all of those VLANs are available for user configuration except for VLAN 4094. All VDCs inherit the new reserved range of VLANs. |

The **VLAN Trunking Protocol (VTP)** is used to create, manage, and maintain a large network with many interconnected switches. The VLAN Trunking Protocol (VTP) can manage the addition, deletion, and renaming of VLANs from a central point without manual intervention and VLAN Trunk Protocol (VTP) thus reduces network administration in a switched network.

## Task 1: Implement the following topology in Packet Tracer.



**\*\*Addressing to be followed, as per the diagram**

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## Task 2: Privilege and Global Configuration Modes.

Privileged mode allows users to view the system configuration, restart the system, and enter router configuration mode. Privileged mode also allows all the commands that are available in user mode. Privileged mode can be identified by the # prompt following the router name. From the user mode, a user can change to Privileged mode, by running the "enable" command.

**Router>enable**

**Router#**

Global Configuration mode mode allows users to modify the running system configuration. From the Privileged mode a user can move to configuration mode by running the "configure terminal" command from privileged mode. To exit configuration mode, the user can enter "end" command or press Ctrl-Z key combination.

**Router# Configure Terminal**

**Router(Config)#**

## Task 3: Setting up VTP Mode and Domain.

### VTP Modes

You can configure a switch to operate in any one of these VTP modes:

- **Server:** In VTP server mode, you can create, modify, and delete VLANs and specify other configuration parameters, such as VTP version and VTP pruning, for the entire VTP domain. VTP servers advertise their VLAN configuration to other switches in the same VTP domain and synchronize their VLAN configuration with other switches based on advertisements received over trunk links. VTP server is the default mode.
- **Client:** VTP clients behave the same way as VTP servers, but you cannot create, change, or delete VLANs on a VTP client.
- **Transparent:** VTP transparent switches do not participate in VTP. A VTP transparent switch does not advertise its VLAN configuration and does not synchronize its VLAN configuration based on received advertisements, but transparent switches do forward VTP advertisements that they receive out their trunk ports in VTP Version 2.

A VTP domain is a set of trunked switches with the matching VTP settings (the domain name, password and VTP version). All switches inside the same VTP domain share their VLAN information with each other.

### Switch 1

Switch>enable

Switch#configure terminal

Switch(config)#hostname Switch1

Switch1(config)#vtp mode server

Switch1(config)#vtp domain Bennett

Changing VTP domain name from NULL to Bennett

Switch1(config)#vtp password Times

Setting device VLAN database password to Times

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## Switch 2

```
Switch>enable
Switch#configure terminal
Switch(config)#hostname Switch2
Switch2(config)#vtp mode client
Setting device to VTP CLIENT mode.
Switch2(config)#vtp domain Bennett
Changing VTP domain name from NULL to Bennett
Switch2(config)#vtp password Times
Setting device VLAN database password to Times
```

## Switch 3

```
Switch>enable
Switch#configure terminal
Switch(config)#hostname Switch3
Switch3(config)#vtp mode client
Setting device to VTP CLIENT mode.
Switch3(config)#vtp domain Bennett
Changing VTP domain name from NULL to Bennett
Switch3(config)#vtp password Times
Setting device VLAN database password to Times
```

### Task 4: VLAN Tagging with Dynamic Trunking Protocol (DTP).

Dynamic Trunking Protocol is used for negotiating a trunk link between two switches as well as the encapsulation type of either 802.1q or ISL.

The different options available while configuring a switch interface:-

**Access:** This mode puts the switch interface into permanent non-trunking mode regardless of whether the neighbouring interface is a trunk port or trying to become a trunk port that is why it is known as DTP mode OFF. The port is a dedicated layer 2 access port.

**Trunk:** It puts the interface into trunking mode. The interface will become a trunk interface even if the neighbouring ports are trunk or not that is why it is called DTP mode ON.

**Dynamic Auto:** This is a default mode on the older CISCO switches. This mode makes the interface able to convert to a trunk link. The interface will become a trunk link if the neighbouring interface is set to trunk or desirable mode. If both switches interface mode is auto, then the trunk will not be formed.

**Dynamic Desirable:** The interface will actively attempt to convert the link into a trunk link. The interface will become a trunk port if the neighbouring interface is set to trunk, desirable or auto.

**Nonegotiate:** This mode prevents the interface from generating DTP frames. This command is used only when the switch port mode is access or trunk. You must manually configure the neighbouring interface as a trunk interface to establish a trunk link.

**\*\*The following table must be consulted for setting up switchport modes.**

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| Administrative Mode    | Auto               | Desirable          | Trunk (on)         | Access (off)       | Non-Negotiate (access) | Non-Negotiate (trunk) |
|------------------------|--------------------|--------------------|--------------------|--------------------|------------------------|-----------------------|
| Auto                   | Static access (NT) | Trunk              | Trunk              | Static access      | Static access          | Unexpected Results    |
| Desirable              | Trunk              | Trunk              | Trunk              | Static access      | Static access          | Unexpected Results    |
| Trunk (on)             | Trunk              | Trunk              | Trunk              | Unexpected Results | Unexpected Results     | Trunk                 |
| Access (off)           | Static access      | Static access      | Unexpected Results | Static access      | Static access          | Unexpected Results    |
| Non-Negotiate (access) | Static access      | Static access      | Unexpected Results | Static access      | Static access          | Unexpected Results    |
| Non-Negotiate (trunk)  | Unexpected Results | Unexpected Results | Trunk              | Unexpected Results | Unexpected Results     | Trunk                 |

```
Switch1(config)#interface fastethernet 0/3
Switch1(config-if)#switchport mode trunk
Switch1(config-if)#interface fastethernet 0/4
Switch1(config-if)#switchport mode trunk
Switch1(config-if)#interface fastethernet 0/5
Switch1(config-if)#switchport mode trunk
```

```
Switch2(config)#interface fastethernet 0/3
Switch2(config-if)#switchport mode trunk
Switch2(config-if)#interface fastethernet 0/4
Switch2(config-if)#switchport mode trunk
Switch2(config-if)#interface fastethernet 0/5
Switch2(config-if)#switchport mode trunk
Switch2(config-if)#interface fastethernet 0/6
Switch2(config-if)#switchport mode trunk
```

```
Switch3(config)#interface fastethernet 0/3
Switch3(config-if)#switchport mode trunk
Switch3(config-if)#interface fastethernet 0/4
Switch3(config-if)#switchport mode trunk
```

### Task 5: Setup and Configure VLAN.

```
Switch1(config)#vlan 10
Switch1(config-vlan)#exit
Switch1(config)#vlan 20
Switch1(config-vlan)#exit
Switch1(config)#interface fastethernet 0/1
Switch1(config-if)#switchport access vlan 10
Switch1(config-if)#interface fastethernet 0/2
Switch1(config-if)#switchport access vlan 20
```

```
Switch2(config-if)#interface fastethernet 0/1
Switch2(config-if)#switchport access vlan 10
Switch2(config-if)#interface fastethernet 0/2
Switch2(config-if)#switchport access vlan 20
```

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```
Switch3(config-if)#interface fastethernet 0/1
Switch3(config-if)#switchport access vlan 10
Switch3(config-if)#interface fastethernet 0/2
Switch3(config-if)#switchport access vlan 20
```

## Task 6: Inter-VLAN Routing.

VLAN traffic between switches we must use a trunk. A trunk connection is simply said nothing more but a normal link, but it can pass traffic from different VLANs and has a method to separate traffic between VLANs. By using trunks, we can make sure all VLAN traffic can be sent between the switches. Because our regular Ethernet frames don't have anything to show to which VLAN they belong we will need another protocol.

There are two trunking protocols:

- **802.1Q:** This is the most common trunking protocol. It's a standard and supported by many vendors.
- **ISL:** This is the Cisco trunking protocol. Not all switches support it.

```
Router(config)#interface fastethernet 0/0
Router(config-if)#no ip address
Router(config-if)#no shutdown
Router(config-if)#interface fastethernet 0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.1.254 255.255.255.0
Router(config-subif)#exit
Router(config)#interface fastethernet 0/0.20
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.2.254 255.255.255.0
```